



GEHealthcare

Technical Publications

GEHC DVMR LCS7397 Liquid Cooling System

**GE Healthcare Part Number: 5161931
Lytron Manual Number 820-0190, Rev. D**

Installation & Service Manual



1.0 SAFETY REQUIREMENTS

WARNING

This cooling system is designed to provide fluid cooling only as specified in this manual. If this cooling system is used in a manner other than as specified, the safety protection of the system may be impaired.

Warnings are posted throughout the manual. Read and follow these important instructions. Failure to observe these instructions might impair safety protection, void the warranty, and can result in permanent damage to the unit, significant property damage, personal injury and/or death.

Be sure to read, understand, and follow all instructions and safety precautions listed in this manual before operating the unit. If there are any questions concerning the operation of the unit or the information in this manual, please contact our Applications Engineering Department at (781) 933-7305.

- **Never place the unit in a location where excessive heat, moisture, or corrosive materials are present.**
- **The unit must be properly grounded at the power source.**
- **Do not connect the SUPPLY or RETURN fitting to your building potable water supply or any pressurized source.**
- **Performance of installation, operation, or maintenance procedures other than those described in this manual or the related service manual may result in a hazardous situation and may void the Lytron warranty.**
- **Transport the unit with care. Sudden jolts or drops can damage the unit.**
- **Observe all warning labels. Never remove warning labels.**
- **Do not operate damaged or leaking equipment.**
- **Do not operate the unit without fluid in the reservoir.**
- **Always turn the unit "OFF" and properly isolate or disconnect cabinet power before performing any service, maintenance procedures, or before moving the unit.**
- **Do not operate equipment with damaged power cords.**
- **A qualified technician should perform service and repairs.**
- **This is not an outdoor cooling system. It must be operated inside a building.**

2.0 CUSTOMER SERVICE SUPPORT

24 hours / 7 days service hotline

Lytron has 24 hours per day, 7 days per week service hotline to help you with questions on the startup and operation of your LCS7397 liquid cooling system.

Lytron is committed to servicing the customer, both during and after the sale.

If you have any questions concerning the operation of your unit, please contact:

LYTRON INC.
55 Dragon Court
Woburn, Massachusetts 01801-1039
(781) 933-7305

To facilitate your call, please have the **model number** and **serial number** of the unit available at the time of the call. The model number and serial number can be found on the Lytron Nameplate, located on the floor of the cabinet, in the lower left hand corner.

Safety Requirements

All units returned for warranty claims must have an RMA (Returned Material Authorization) number on the outside of the container. Call Lytron Customer Service at (781) 933-7305 for an RMA number. Refer to the end of manual for the system warranty.

3.0 COMPLIANCE STANDARDS

The DVMR LCS7397 is in agreement with the following standards:

US/CAN

Safety

US - UL 61010A-1 - Laboratory & Testing and Measurement Equipment

CAN - CSA 22.2, No. 1010.1 - Laboratory & Testing and Measurement Equipment

EMC

FCC Part 15 - Unintentional Radiators

CE/EU

Safety

EN61010-1- Laboratory & Testing and Measurement Equipment

EMC

EN61326 - Radio and Conductive Emissions



4.0 DECLARATION OF CONFORMITY



DECLARATION OF CONFORMITY

Product Description: Liquid Cooling System

Model Number: LCS7397G3

We, Lytron Incorporated, 55 Dragon Court, Woburn, Massachusetts, 01801, USA, declare that the product described above is in conformity with the relevant EU Directives listed below using the relevant section of the following EU standards and other normative documents.

LOW VOLTAGE DIRECTIVE 2006/95/EC

Essential health and safety requirements relating to electrical equipment designed for use within certain voltage limits

EN61010-1:2001

Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use - Part 1: General Requirements

EMC DIRECTIVE 2004/108/EC

Essential health and safety requirements relating to electromagnetic compatibility

EN 61326:1997/A1:1998/A2:2001/A3:2003

Class A, Conducted Emissions, 150kHz to 30MHz
Class A, Radiated Emissions, 30MHz to 1GHz

EN 61326:1997/A1:1998/A2:2001/A3:2003

EN61000-4-2:1995/A1:1998/A2:2001 – Electrostatic Discharge, 8kV Direct Air, 4kV Direct and Indirect Contact
EN61000-4-3:2002/A1:2002 – Radiated RF Immunity, 80 MHz-1000MHz, 10V/m, 80% AM 1kHz
EN61000-4-4: 2004 – EFT, 2kV Direct to Power Leads, 1.0 kV to I/O
EN61000-4-5:1995/A1: 2001 – Surge Immunity, 2kV Common Mode
EN61000-4-6:1996/A1: 2001 – Conducted RF Immunity, 150kHz-80MHz, 3Vrms, 80% AM 1kHz
EN61000-4-11:2004 – Voltage Dips and Interruptions

This product has been qualified in accordance with the requirements established above and is manufactured in compliance with Lytron's registered EN ISO 9001:2000; BS EN ISO 9001:2000; ANSI/ASQ Q9001:2000 Quality System.


Authorized Lytron Representative

10/6/08
Date

55 Dragon Court / Woburn, MA 01801 USA / Tel 781-933-7300 / Fax 781-935-4529 / www.lytron.com



F4.03.36 Rev A 10/02/08

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Revision History

REV	REASON FOR CHANGE	APPROVED BY / DATE
A	Initial Release	A. Lupien 10/20/06
B	Revised per ECN# 15890	N. Grundy 10/08/08
C	Revised pages15, 17 per ECN# 16335	N. Grundy 04/10/09
D	Revised pages 15, 23 per ECN# 16391	N. Grundy 5/12/09

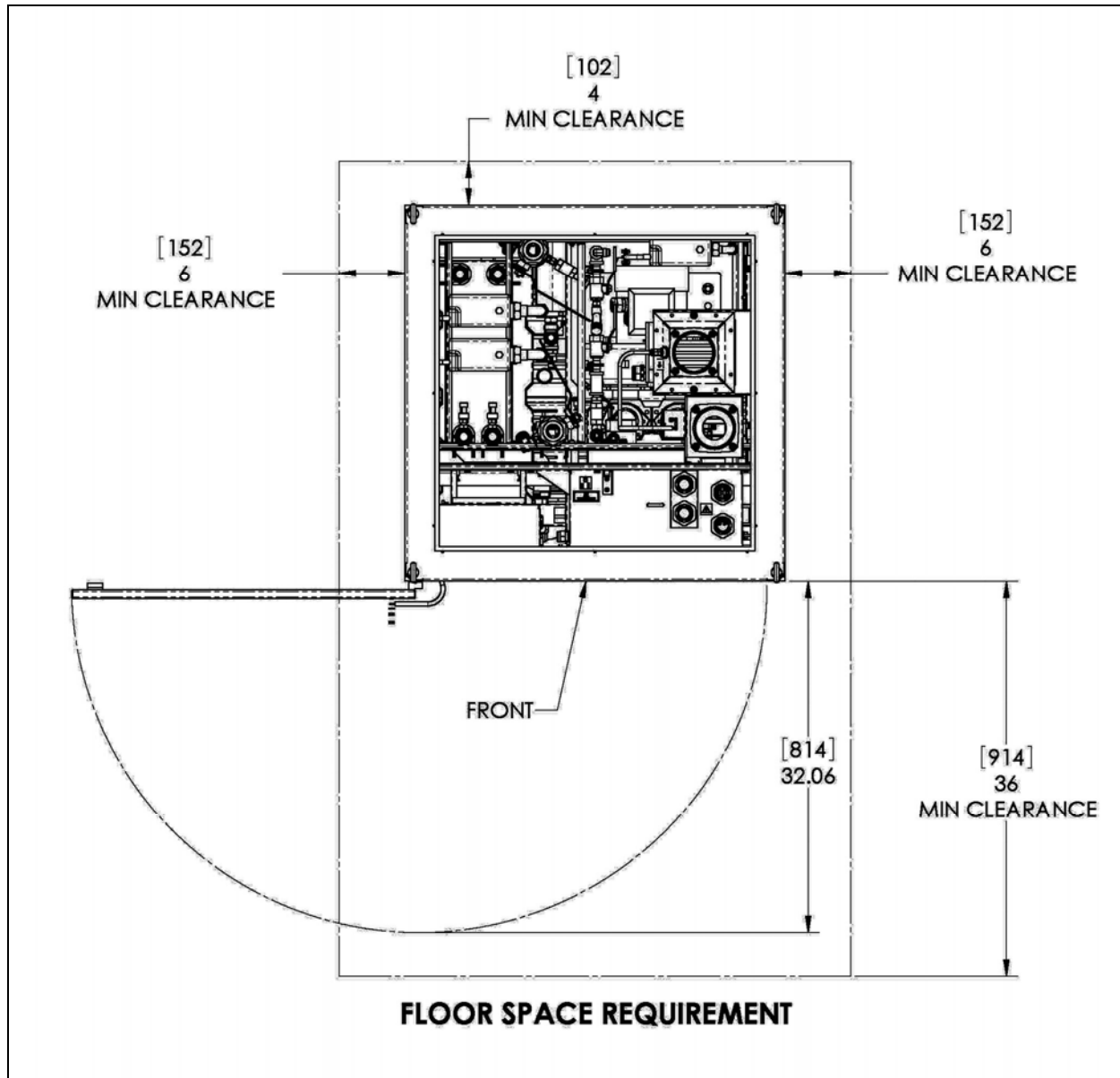
5.0 LCS7397 SPECIFICATIONS

Subject	Specifications
Overall Dimensions	35" (89 cm) wide x 35" (89 cm) deep x 75" (191 cm) high
Shipping Crate Dimensions	48" (122 cm) wide x 48" (122 cm) deep x 83" (211 cm) high
Weight (empty, no fluid)	1000 pounds (454 kg)
Shipping Weight (empty, no fluid)	1085 pounds (492 kg)
Center of Gravity (empty, no fluid)	Approximately 48" (122 cm) from floor, centered in cabinet.
GC Tank Volume	18 gallons (68 liters)
PE Tank Volume	18 gallons (68 liters)
GC Pump Capacity	7.1 GPM (27 L/min) @ 58 PSI (4 bar) @ 60 Hz 7.1 GPM (27 L/min) @ 58 PSI (4 bar) @ 50 Hz
PE Pump Capacity	30 GPM (113.5 L/min) @ 60 PSI (4.14 bar) @ 60 Hz 30 GPM (113.5 L/min) @ 60 PSI (4 bar) @ 50 Hz
Recommended Process Fluid – Gradient Coil & Power Electronics	GC Circuit – DI water with resistivity between 1 and 2 Mohm (at 25° C) PE Circuit – DI water with resistivity between 1 and 3 Mohm (at 25° C)
Recommended Facility Fluid	Water with 0% to 40% propylene or ethylene glycol by volume for intake by the LCS7397 under the following conditions: PH: 6.5 to 9 Hardness: Less than 200 PPM calcium carbonate Suspended Matter: <10 mg/L; 150-micron particle size
Process Fluid Connections	GC Supply - 1" SAE 37/45°. JIC flare GC Return - 1" SAE 37/45°. JIC flare PE Supply - 1" SAE 37/45°. JIC flare PE Return - 1" SAE 37/45°. JIC flare
Facility Fluid Connections	Facility Supply – 1 ½" NPTF Facility Return – 1 ½" NPTF Cryo Supply – ½" SAE 37°. JIC flare Cryo Return – ½" SAE 37°. JIC flare
Electrical Requirements	480 volt (+/- 10% daily variation; +/- 14% up to 200 msec.), 3 phase, 60 Hz (+/- 3 Hz variation) 380 volt (+/- 10% daily variation; +/- 14% up to 200 msec.), 3 phase, 50 Hz (+/- 3 Hz variation)
Recommended Circuit Breaker	40 amps minimum
Ambient Temperature – Non-Operating	-40 to 70° C with a maximum positive/negative rate of change of 20° C/hr
Humidity – Non-Operating	Between 5% and 100%, condensing, with a positive/negative rate

	of change of 30%/hr
Altitude – Non-Operating	30.5 m below sea level to 5200 m above sea level. Non-operating conditions cover air transport of the cooling cabinet
Magnetic Field – Non-Operating	30 Gauss
Ambient Temperature – Operating	10 to 35° C with a maximum positive/negative rate of change of 10° C/hr. Maximum allowable temperatures may be reduced for high altitude operation (greater than sea level) by a factor of 1.5 deg. C per 1000 m.
Humidity – Operating	Between 30% and 75%, non-condensing, with a maximum allowable positive/negative rate of change of 5%/hr
Altitude – Operating	30.5 m below sea level to 2500 m above sea level. Non-operating conditions cover air transport of the cooling cabinet
Magnetic Field – Operating	30 Gauss
Elevation Above MR System	5 m
Elevation Below MR System	5 m
Power Cord Requirement for mains power	Temperature Rating: 65° C minimum Current Rating: 35Amp minimum.

6.0 LCS7397 DIMENSIONS

6.1 Overall Dimensions



Detail from 200-2243, Rev E

6.2 PRODUCT IDENTIFICATION NAMEPLATE

SUPPLIER CODE XXXXXXXXXX		LYTRON Fluid Thermal Solutions		PRODUCT OF USA 09/17/08	
66 DRAGON COURT, WOBURN, MA 01801-1039 For Service and Support World Wide Call: 781-933-7300 Or E-mail: service@Lytron.com					
MODEL NO.			SERIAL NO.		
			XXXXXXXXXX		
VOLTS	HZ	PHASE	AMPS	Maximum Breaker Rating 50A	
PUMP TYPE		GEHC P/N & REV			
		X			
CERTIFICATION OF COMPLIANCE					
Approved by:					

The Product ID Label identifies the Lytron model number and serial number, GEHC part number and latest revision, electrical information, and pump type. This label is located on the inside floor of the cabinet in the front side left hand corner.

7.0 SAFETY LABELS AND MARKINGS



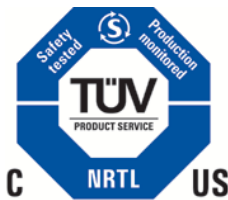
The Protective Earth (ground) Terminal.

La borne de mise a la masse



This symbol certifies that the system conforms to the EMC Directive and all applicable directives at the time of marking.

Ce symbole certifie que le refroidisseur de liquide conforme a la directive EMC et a tous directives a l'heure de l'inscription.



Agency compliance and approval product conforms to standards.

La conformite d'agence et l'approbation de produit se conforme aux normes



Attention: This symbol tells the user to consult accompanying documents before servicing equipment.

Ce symbole indique a l'utilisateur de consulter des documents d'accompagnement avant d'enteretenir l'equipment.



High Voltage:

Disconnect power before servicing.

Debrancher toute alimentation electrique avant entretien



Frequency Toggle Switch, 380V~50Hz and 480V~60Hz

Interrupteur a levier de frequence

8.0 RECEIVING & UNPACKING THE LCS7397

8.1 Receiving the LCS7397

- 8.1.1 The LCS will be shipped from Lytron on a wooden pallet. This pallet and packaging material is designed to protect the cooling system from potential damage during transit. The shipping pallet is a custom wooden pallet that has been specifically engineered for the LCS7397. The LCS will be bolted to the pallet via metal angle mounting brackets. All eight edges of the cabinet will be protected by cardboard angle and the entire cabinet shrink-wrapped. In addition, the cabinet will be banded to the pallet in 2 places.
- 8.1.2 The pallet is approximately 44" (112 cm) wide x 45" (114 cm) deep x 75" (190 cm) high. The overall height (unit on pallet) is approximately 80" (203 cm). See section 6.0 of this manual for more dimensional information.
- 8.1.3 The LCS can be moved with a pallet jack with a capacity of ≥ 1500 lbs (500 kg) or with a fork truck with a capacity of ≥ 1500 lbs (500 kg). The forks should be centered underneath the packaging pallet.
- 8.1.4 Unpacking the LCS7397
- 8.1.5 Begin unpacking the LCS by cutting the two nylon straps that fasten the cabinet to the pallet. Remove the shrink-wrap and cardboard angles that surround the cabinet. Discard the nylon straps, shrink-wrap, and cardboard angles.
- 8.1.6 Remove the two metal angle-mounting brackets from the LCS base. The two metal angle-mounting brackets can remain screwed to the pallet.

WARNING: The LCS should not be moved while still positioned on pallet if the metal angle mounting brackets have been removed.

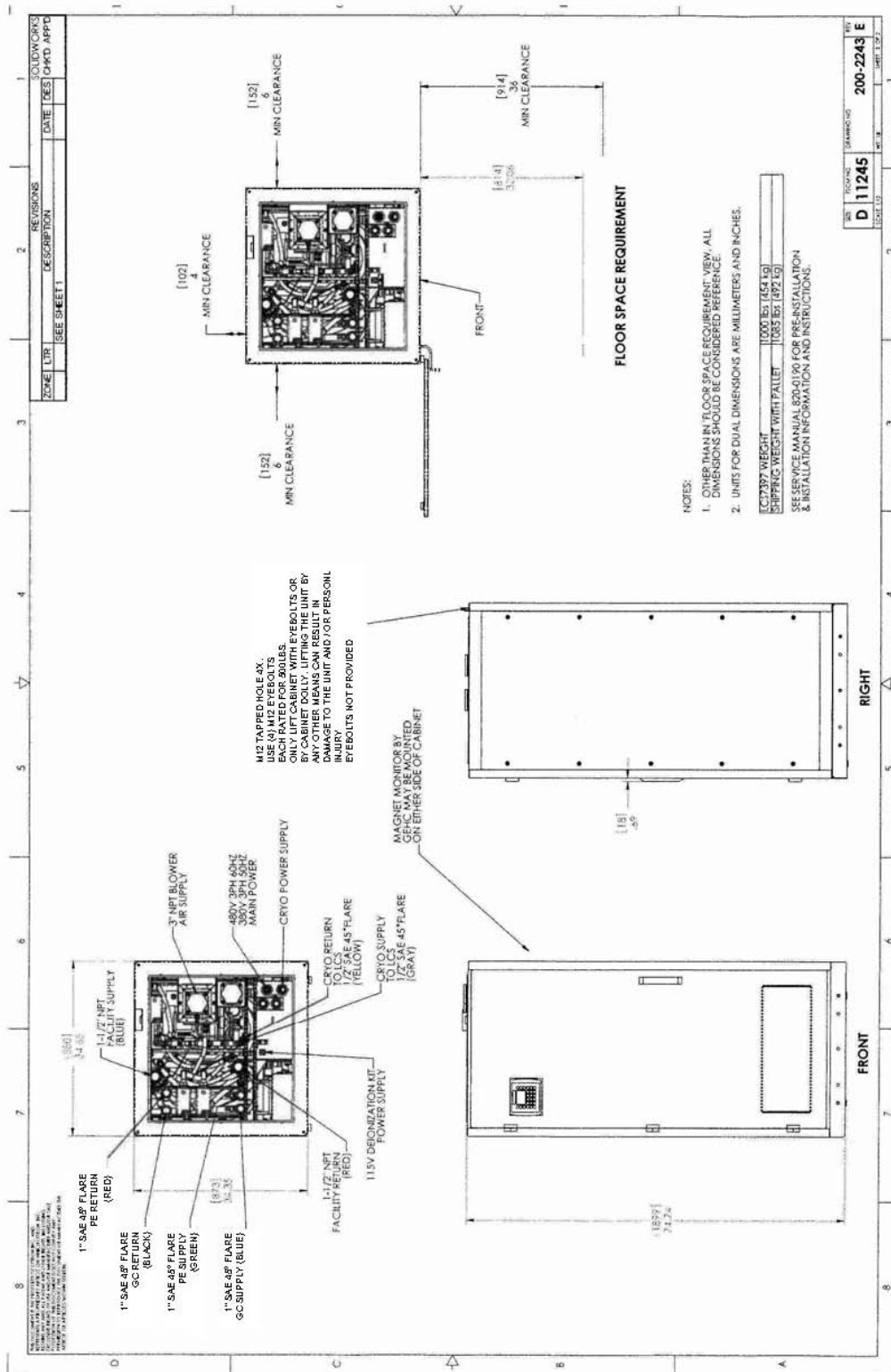
- 8.1.7 Removing the LCS from pallet: There are (4) $\frac{1}{2}$ -13 tapped holes provided on the top of the cabinet in each of the four corners. These holes are designed to accept a $\frac{1}{2}$ -13 eyebolt. It is intended that the cabinet be lifted via eyebolts. The LCS should only be lifted using 4 eyebolts; each rated for 500 lbs (227 kg) minimum. Lifting method and mechanism to be determined by customer. **Note: Eyebolts will not be provided with LCS.** An alternate method for removing the LCS from the pallet is to use the Cabinet Dollies. See the installation manual for details.
- 8.1.8 In determining the final location for LCS, follow the recommendations provided in section 10.3 of this manual.

9.0 FACILITY REQUIREMENT

Ambient Conditions and LCS location

- 9.1.1 Ambient Conditions: The local ambient temperature should be between 10 to 35° C with a maximum positive/negative rate of change of 10° C/hr. Maximum allowable temperatures might be reduced for high altitude operation (greater than sea level) by a factor of 1.5 deg. C per 1000 m. The humidity level should be between 30% and 75%, non-condensing, with a maximum allowable positive/negative rate of change of 5%/hr. The LCS should not be operated at an elevation greater than 16.5 ft (5m) above or below the MR system
- 9.1.2 Location: The LCS7397 should be installed in a location that allows airflow around the unit. The LCS must have at least 6 inches (13 cm) of clearance at the rear of the cabinet and 36 inches (91 cm) of clearance at the front of the cabinet (the front is defined as the side with the hinged door. The rear is opposite the front side). This space must be provided for proper airflow. Space above the LCS should be considered for all electrical and plumbing connections.

9.1.2.1 Floor Space Diagram



10.0.0 Electrical Requirements

- 10.0.1 Fused Circuit- The LCS7397 must be supplied with a dedicated, fused circuit. The circuit breaker should be 40-amp minimum.
- 10.0.2 Power- Power to the LCS7397 shall be received from the DVMR main disconnect panel. The LCS7397 requires 480 volt (+/- 10% daily variation; +/- 14% up to 200 msec.), 3 phase, 60 Hz (+/- 3 Hz variation) or 380 volt (+/- 10% daily variation; +/- 14% up to 200 msec.), 3 phase, 50 Hz (+/- 3 Hz variation)
- 10.0.3 The power cord delivering power from the DVMR main disconnect panel to the mains terminal blocks in the LCS7397 electrical box must be rated as follows:
 - Temperature Rating: 65° C minimum
 - Current Rating: 35Amps minimum
- 10.0.4 Ground- a main power ground stud is provided inside of the electrical box, adjacent to the main power terminal strip. An appropriate ground wire must be installed to the ground stud provided.

10.1 Plumbing

- 10.1.0 Refer to section 6.0 for a call out of all required plumbing connections.
- 10.1.1 When making plumbing connections to Parker SAE 45° flare fittings, manufacturer assembly recommendations should be followed. As a general rule, Parker recommends hand tight, then 1/4-turn with a wrench for their flare connections.
- 10.1.2 When making plumbing connections to NPT fittings, Lytron recommends Real-tuff (Hercules Chemical Company, Inc.) for use as a thread sealant. Follow the Real-tuff application recommendations.

10.2 Facility Fluid (customer supplied coolant)

- 10.2.1 The customer supplied coolant must meet the following conditions:
- 10.2.2 Water with 0% to 40% propylene or ethylene glycol by volume for intake by the LCS7397.
- 10.2.3 PH: 6.5 to 9
- 10.2.4 Hardness: Less than 200-PPM calcium carbonate
- 10.2.5 Suspended Matter: <10 mg/L; 150-micron particle size
- 10.2.6 Minimum flow rate of 114 L/min (30 GPM) for 40% propylene glycol water coolant.

10.2.7 Maximum facility coolant supply pressure shall be 6 bar (87 PSI) measured at the inlet of the cooling cabinet.

10.2.8 The maximum facility coolant temperature is 10° C (52° F) for 40% propylene glycol.

10.2.9 The minimum facility coolant temperature is 0° C (32° F).

10.3 Process Fluid Requirements (process water for GC & PE plumbing circuit)

10.3.1 There are two independent process water circuits that are designed to cool the Gradient Coil and the Power Electronics. Each requires slightly different water specifications. Below is the specification for each.

10.3.1.1 (GC) Gradient Coil Circuit – DI water with resistivity between 2.5K Ohm/cm-1000K Ohm/cm (at 25° C)

10.3.1.2 (PE) Power Electronics Circuit – DI water with resistivity between 2.5K Ohm/cm-1000K Ohm/cm (at 25° C)

11.0 TRANSPORTING REQUIREMENTS

11.1 Draining Procedure

WARNING

Always disconnect power to LCS when performing service.

Not following this procedure may result in extensive damage to the LCS. It is extremely important to completely drain the LCS when not in use for extended period of time, shipping outside of its current location, during certain repairs, and for scheduled fluid maintenance. Lytron recommends method #1 as the most efficient way to remove fluid from the cooling system.

ATTENTION

A container should be available to hold the fluid removed from the LCS. This container should be labeled in accordance with OSHA rules and regulations concerning handling of the fluid being removed from the LCS.

Reference the MSDS sheet for the fluid being removed from the LCS for further handling, storage, and disposal instructions. Disposal of fluid removed from the Lytron cooling system should be in accordance with all Local and Federal disposal laws.

There are 6 drain ports located on the LCS7397 cooling system. Each of the 6 connections will have to be utilized to properly drain the system.

DRAIN PORTS



Figure 1

Figure 1 shows both the heat exchanger drain Quick Disconnect (QD) and the tank drain. Each of the two heat exchangers and each of the two tanks have a drain.

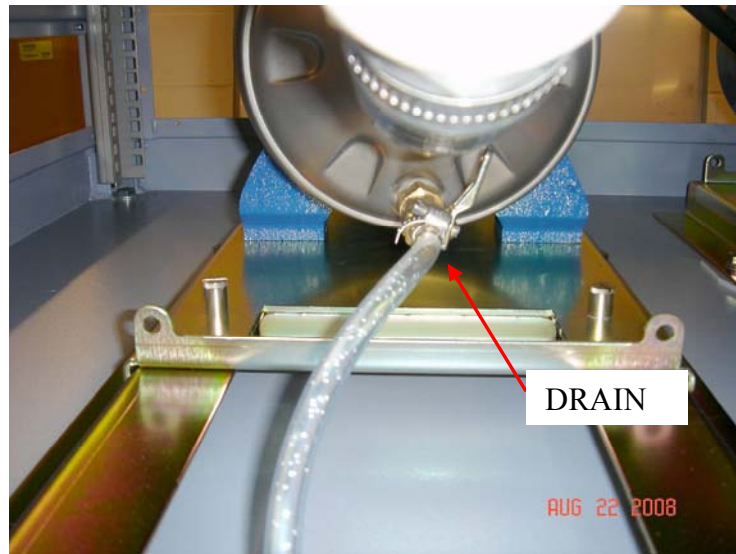


Figure 2

Figure 2 shows the pump drain. Each of the two pumps has a drain.

11.1.2 To drain the heat exchangers and the tanks you will need the drain hose supplied with the water removal kit, 5269683, and the approved container for the fluid.

11.1.3 To Drain the Pump-only circuit

- Shut off gate valve at bottom of reservoir
- Unwind hose from located at bottom of pump – direct hose into suitable container
- Open ball valve located at the pump
- Remove the bolt on the face of the pump 180° from the drain valve

11.1.4 To Drain Facility plumbing circuit:

- 11.1.5 Before draining the facility plumbing circuit, the facility water supply must be isolated from the cooling system. To isolate the facility water supply, close the 1 ½” ball valves on the facility supply and return headers. Disconnect the facility supply and return lines from the LCS.
- 11.1.6 Facility water is drained from the LCS through the two brass quick disconnects located on the bottom of the PE and GC heat exchangers (figure 1).
- 11.1.7 Important: Determine where the fluid from the LCS will be contained prior to attaching the drain hose to the QD’s.
- 11.1.8 Using the drain hose described in 11.1.2, connect to the brass QD fitting on either of the two heat exchangers. Crack open the 1 ½” ball valve on the facility supply and return line to vent the headers. The facility fluid will be sufficiently drained when fluid stops running from drain hose. After one heat exchanger is drained, move drain hose to the other heat exchanger and allow to drain.
- 11.1.9 To Drain GC & PE plumbing circuit:
- The PE and GC plumbing circuits are each drained through two places, the tank and the pump. The tank drain is a quick disconnect fitting located on the right side of each tank. The pump drain is a ball valve with 3’ of clear tubing located at the bottom of each pump housing.
- 11.1.10 Remove the tank fill cap from both tanks to allow tank to vent.
- 11.1.11 To drain the GC plumbing circuit, begin by first removing the bulk of the volume from the tank. Attach the drain hose to the quick disconnect on the side of the tank (figure 1). The tank will be sufficiently drained when fluid stops running from drain hose. Remove the drain hose from the QD.
- 11.1.12 Drain the pump and remaining fluid from the pump suction line by opening the ball valve located on the bottom of the pump housing (figure 2). The GC plumbing circuit will be sufficiently drained when fluid stops running from pump drain hose. Close the pump drain ball valve when fluid stops running from the hose.
- 11.1.13 To drain the PE plumbing circuit, perform the same steps as described for the GC draining procedure.

12.0 SHIPPING

Whenever the cooling system is going to be transported it should be packaged on the original shipping skid. The following should be followed when shipping the LCS7397:

Center the LCS on the skid between the foam supports. The LCS should be bolted to the skid via the angle mounting rails as shown in the photo:



After bolting the cabinet to the skid, 4 vertical corners and 4 horizontal corners at the top of the cabinet should be lined with angle cardboard for protection. The entire cabinet can be wrapped with plastic shipping wrap as shown in photo below:



In addition, the LCS should be banded to the skid by running shipping band under the skid and over the cabinet in two places, opposite the metal angle mounting rails as shown in the photo above.

This skid is designed to protect the cooling system from potential damage during transit. The shipping crate is a custom wooden pallet that has been specifically engineered for the LCS7397. The pallet is approximately 44" (112 cm) wide x 48" (114 cm) deep x 80" (203 cm) high. If the original packaging is unavailable, contact Lytron's Service Department for shipping instructions prior to shipping this unit.

WARNING: Shipping this product using anything other than its original packaging can result in personal injury.

13.0 COOLING SYSTEM START-UP AND OPERATION

13.1 Electrical Power

13.1.1 The electrical power requirements are specified in section 10.1 of this manual.

13.1.2 Prior to hooking up the main power, verify that the 5 circuit breakers are in the “DOWN” position.

13.1.3 Main power will be supplied, by facility, to the cabinet through the strain relief located in the top of the LCS electrical box. The three power wires should be brought to the terminal strip inside the electrical box and connected to the terminal strip as shown on the Electrical Wiring Diagram (Lytron Drawing #830-0171). There is a ground stud provided adjacent to the terminal strip for the ground lead.

Prior to starting the LCS it is extremely important to identify what voltage the LCS will be run at (380V or 480V). See Table 1 below. Figure 3 shows the 380/480V switch.

TABLE 1	
Nominal Input Voltage (VAC)	HEC Switch Position
380V	380V
400V, 415V, 480V	480V

WARNING

Failure to switch the 380/480V switch to the correct operating voltage could result in irreversible damage to the LCS control system.



Figure 3

13.1.4 After you have hooked up the main power and positioning the 380/480V switch, install electrical box cover.

13.2 Filling the LCS

13.2.1 The LCS has three separate fluid requirements. Facility Fluid, GC Fluid, and PE Fluid. The specification for each are described in section 10.2 & 10.3.

13.2.2 Direction to fill the LCS:

1. Verify the gate valve located in the suction line of the pump is fully “OPEN”.
2. Verify that both pump drains are “CLOSED”. The pump drain is a ball valve located at the bottom-front of the pump housing.
3. The GC (Gradient Coil) plumbing circuit is located on the left hand side of the cabinet when looking at the cabinet from the hinged door. The PE (Power Electronics) plumbing circuit is on the right hand side of the cabinet. Remove the fill port caps from both tanks.
4. Fill the GC tank with DI water as specified in section 10.3 of this manual. The initial volume required to fill the GC tank will be approximately 18 gallons.
5. Fill the PE tank with DI water as specified in section 10.3 of this manual. The initial volume required to fill the PE tank will be approximately 18 gallons.
6. Replace caps on each tank.

NOTE: More fluid will be required to “top-off” the tank later in the start-up procedure after the pumps have been run and the process fluid headers are filled.

13.3 Powering the Signal Box, GC and PE Pumps, Blower, & Cryo Compressor

13.3.1 There are 5 circuit breakers located in the electrical box. They are shown in the photo below:

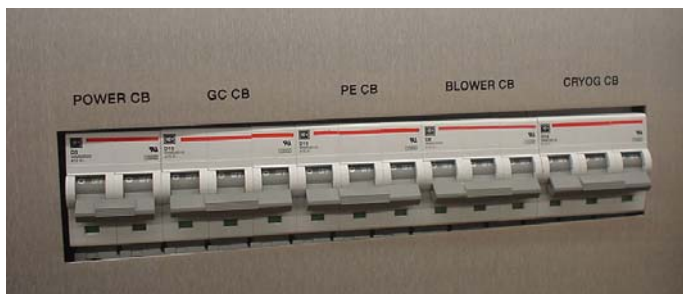


Figure 4

- 13.3.2 To power the devices protected by these circuit breakers, the breaker needs to be switched to the “up” position. The devices protected by the circuit breakers are from left to right – “Power CB” = 115V power to the Signal Box, “GC CB” = power to the GC (Gradient Coil) VFD & pump, “PE CB” = power to the PE (Power Electronics) VFD & pump, “BLOWER CB” = Power to the Blower, “CRYOG CB” = power to the Cryo Compressor.

Note: Switching the circuit breakers to the “ON” position will not start the devices protected by the breakers.

13.3.3 Starting Pumps

- 13.3.4.1 Operation of the GC pump, PE pump, Blower, and Cryo Compressor will be controlled by GE protocol via the host computer. Refer to GE documentation for instructions on controlling these devices.

IMPORTANT – At initial start-up of the GC & PE pumps, volume in the tank will be used to fill the supply & return lines. Follow the instructions below to properly start the pumps.

- 13.3.1.2 The GC and PE pumps should be run at increments of 10 seconds at initial start-up. After each start, stop the pump after 10 seconds of operation. Checking the tank volume after each stop. If tank volume drops, “top off” tank with DI water as specified in 10.4
- 13.3.1.3 After the supply and return lines are full and the pumps are pumping fluid throughout the plumbing circuit the tank volume should remain at a constant level approximately 2 inches from the top of the tank.

14.0 REGULARLY SCHEDULED MAINTENANCE

There are a few things you can do to your LCS to make sure that it will operate at peak efficiency. The following steps should be taken every 6 months or as described below.

14.1.2 Deionizing and Filtering the Cooling system

- 14.1.3 To clean the Process (GC/PE) side of the cooling system and prevent algae growth within the fluid cooling system, GE recommends that the deionization and filtering procedure be followed per the planned maintenance cycle. Refer to GE documentation for instructions on performing the Deionization and Filtering procedure.system Cryo Strainer Replacement

14.1.4 Normal Scheduled Maintenance

Lytron recommends that the mesh filter element of the cryo supply strainer (P/N '5269685) be changed every 6 months as part of a scheduled maintenance program.

14.1.5 Start-up Maintenance

It is recommended that the element be changed after the LCS7397 facility plumbing circuit is allowed to run for 1-2 weeks. This is due to a higher than normal load on the mesh screen due to particulate left in the water circuit at manufacture.

14.1.6 Instructions for Replacing Element:

CAUTION

Disconnect power to cooling systems prior to servicing. Wear latex gloves and safety glasses when performing this service.

NOTE: Disposal of fluids removed from the LCS7397 and related plumbing circuits should be in accordance with all local and Federal disposal laws.

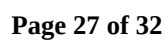
- Step #1 **Note: It is important to first open the bypass ball valve prior to closing the strainer isolation valves to prevent flow from being interrupted to the Cryo Compressor.**
The cryo supply line must be in “by-pass” mode to replace filter mesh. Access to the Cryo supply line will be from the top of the cooling cabinet. To put the Cryo supply line in bypass, first open the ball valve located in the cryo bypass line. Second, close the two ball valves located on either side of the strainer.
- Note: There may be fluid spilled during the mesh strainer replacement. Have a method to capture up to 12 oz. of facility fluid.**
- Step #2 Using adjustable wrench; remove the hex plug on the wye of the strainer. Remove the contaminated mesh screen and dispose. **Do not reuse mesh screen. Contamination may not be clearly visible.**
- Step #3 Install new mesh screen then replace hex cap.
- Step #4 Return Cryo supply to normal operation. First, open both ball valves on either side of strainer. Second, close the ball valve located in the Cryo bypass line.
- Step #5 Dry the wye strainer with paper towels then visually inspect for leaking

15.0 FIELD REPLACEABLE UNITS (FRU'S) & SPARE PARTS

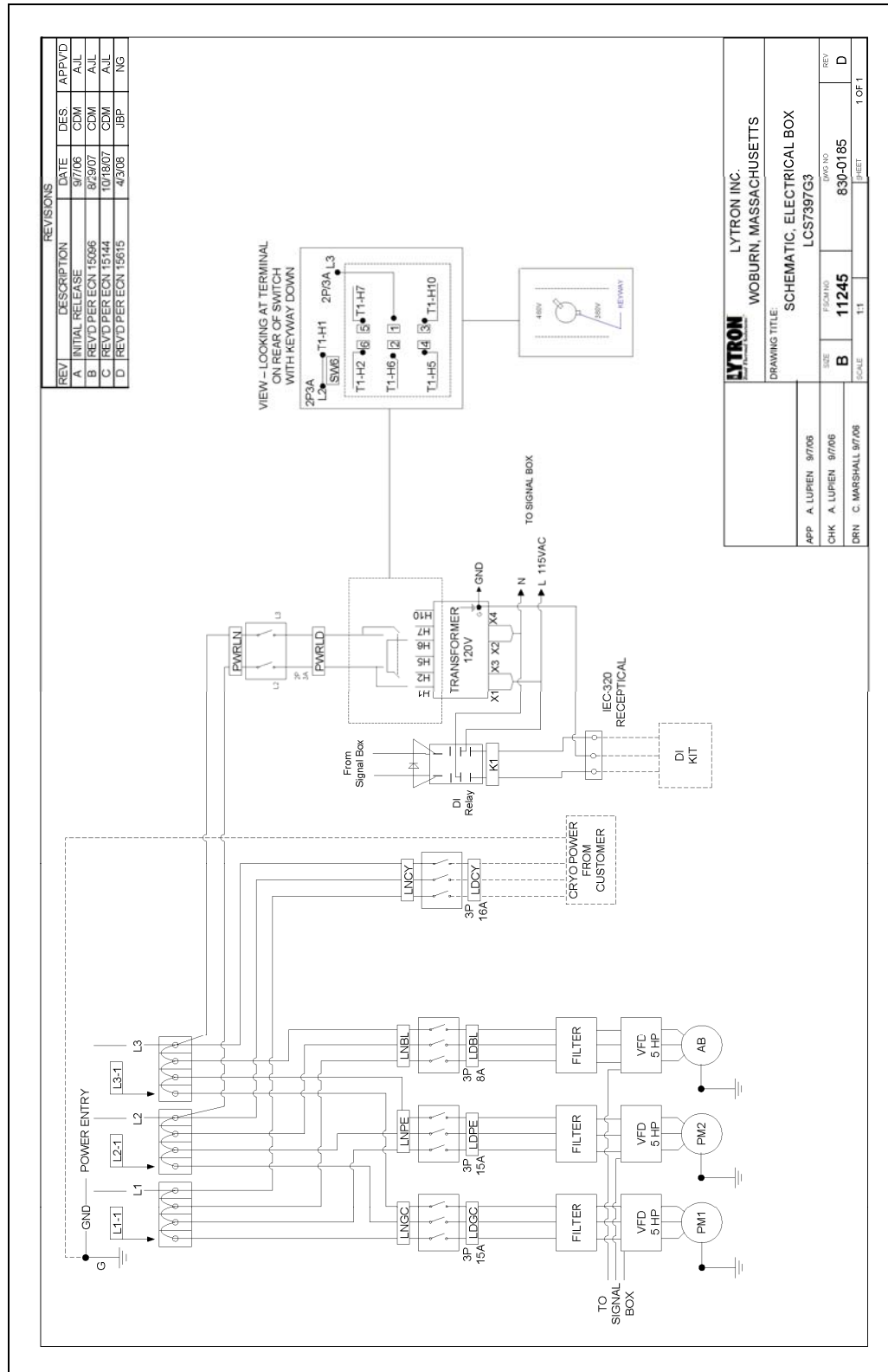
FRU Assemblies

Lytron P/N	GE P/N	Part Description
205-0121	5156316	Cryogen Compressor Flow Meter Kit
205-0121-01	5156321	5hp Pump Kit
205-0121-02	5156318	Mixing Valve Kit for Gradient Coil and Power Electronics Cabinet Cooling Loops
205-0121-03	5156320	Mixing Valve Kit for Chilled Air Blower
205-0121-04	5156315	3hp Blower Kit
205-0121-05	5156319	Power Box Kit
205-0121-06	5269691	Signal Box Kit
205-0121-07	5267643	Liquid Pressure Sensor Kit
205-0121-08	5268665	Liquid Temperature Sensor Kit
205-0121-10	5269689	Air Temperature Sensor Kit
205-0121-11	5267644	Level Switch Kit
205-0121-12	5267649	Resistivity Sensor Kit
205-0121-13	5269685	Cryogen Compressor Filter Kit
205-0121-14	5267648	Pressure Difference Sensor Board Kit
205-0121-15	5269683	Coolant Removal Kit
205-0121-16	5266827	Fru, fuse kit, DI and Transformer
205-0121-17	5262029	Support Tube, HDV EQ Room
MN7519G1	5224090	DVMR DI Kit (Already Quoted. Not a Fru defined in spec)
205-0116	5264701	DVMR DI Kit FRU (Already Quoted. Not a FRU defined in spec. This is replacement part for 5224090)

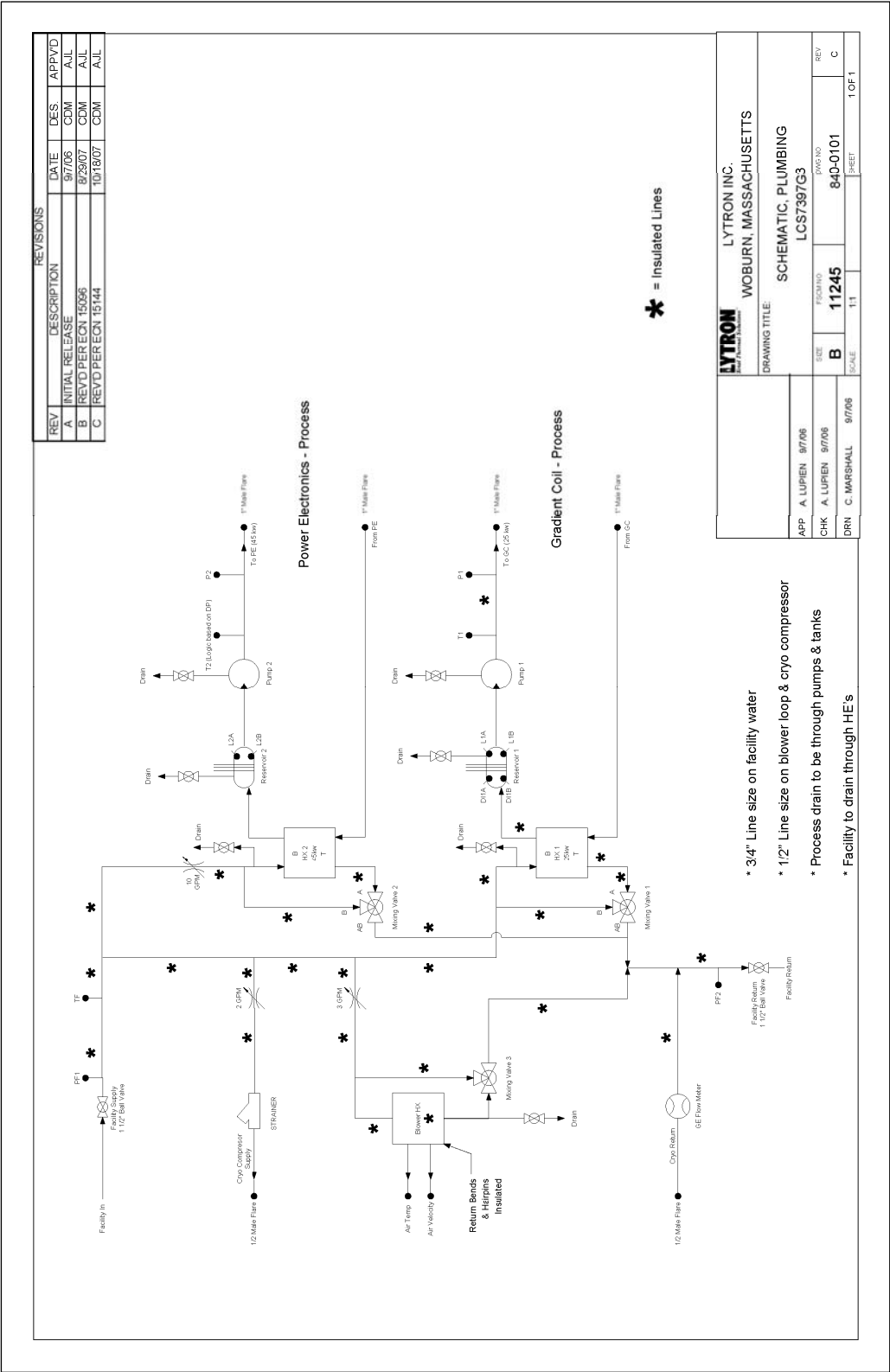
* Part warranty is 90 days or remainder of system, whichever is greater.



17.0



18.0 PLUMBING SCHEMATIC



19.0 SERVICE POLICY

Lytron's cooling systems are the product of over 40 years of thermal engineering and manufacturing experience. We designed them to provide superior reliability, easy maintenance and worry-free operation. However, occasionally a system may need repair. To ensure your process is back up and running quickly, Lytron has implemented the following cooling system service policy.

15.1 Diagnostic Consultation:

At no cost, Lytron will attempt to diagnose the problem over the phone. Our service department can be reached by calling 781-933-7305 and following the menu. Service technicians are available 24 hours/7 days for consultation. Lytron strongly encourages customers to take advantage of this service before returning a cooling system to Lytron for evaluation. Often a problem with a system can be fixed quickly in-house or is an application problem. By working with Lytron's service department to troubleshoot a cooling system, you do not have the downtime and expense associated with returning the system to our factory.

15.2 Replacement Parts:

Replacement parts can be ordered using a credit card or purchase order. Parts being returned from systems under warranty should be returned using a Lytron issued RMA number. If the parts are found to be defective and the claim is within the warranty period, credit will be issued for the price of the parts and one-way ground shipping charges. If the parts are not defective or indicate end user damage, no credit will be issued. Lytron will not cover the incremental cost of air shipment of replacement parts, regardless of warranty status. In-stock parts normally will be shipped the next business day; non-stocked parts will be shipped as quickly as reasonably possible. This policy is subject to change. Please check with Lytron's service department for the current policy.

Ref. Form F4.3.25 Rev D, 7/21/04

20.0 WARRANTY

Lytron's Standard Warranty

Lytron agrees that the apparatus manufactured by it will be free from defects in materials and workmanship for the warranty period under normal use and service and when properly installed. The warranty period for recirculating chillers is two years from date of shipment of such apparatus to the original purchaser, maintenance items excluded, and one year from date of shipment of such apparatus to the original purchaser for all other products Lytron sells. Lytron's obligation under this agreement is limited solely to repair or replacement, at its option, at its factories, of any part or parts thereof, returned to Lytron with transportation charges prepaid, which examination shall disclose to Lytron's satisfaction to have been defective. **THE FOREGOING EXPRESS WARRANTY IS IN LIEU OF ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. LYTRON'S OBLIGATION UNDER THIS WARRANTY IS STRICTLY AND EXCLUSIVELY LIMITED TO THE REPAIR OR REPLACEMENT OF DEFECTIVE COMPONENT PARTS AND LYTRON DOES NOT ASSUME OR AUTHORIZE ANYONE TO ASSUME FOR IT ANY OTHER OBLIGATION. LYTRON ASSUMES NO RESPONSIBILITY FOR INCIDENTAL, CONSEQUENTIAL, OR OTHER DAMAGES INCLUDING, BUT NOT LIMITED TO LOSS OR DAMAGE TO PROPERTY, LOSS OF PROFITS OR REVENUE, LOSS OF THE UNIT, LOSS OF TIME, OR INCONVENIENCE.** Lytron's liability does not include any labor charges for replacement of parts, adjustments, repairs, or any other work done outside its factories and its liability does not include any resulting damage to persons, property, equipment, goods or merchandise arising out of any defect in or failure of its apparatus. Lytron's obligation to repair or replace shall not apply to any apparatus which shall have been repaired or altered outside of its factory in any way, or which has been subject to negligence, to misuse, or to pressures in excess of stated limits. On parts not of Lytron's manufacture, such as motors, controls, etc., Lytron extends only those warranties given to Lytron, Inc. to the extent Lytron can do so. Lytron's agreement hereunder runs only to the immediate purchaser from Lytron, Inc. and does not extend, expressly or by implication, to any other person.

Ref. Form F4.3.18 Rev. E Effective June 6, 2003



GE Healthcare

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